

Project Proposal

Sociocultural theories of learning shift the focus from simply acquiring content to engaging in meaningful activity with others. Instead of viewing knowledge as something transferred to individuals, learning is understood as the development of competence through shared practice within communities (Lave & Wenger, 1991). Gutiérrez and Jurow (2016) build on this idea by highlighting that learners' community histories and funds of knowledge should be central to design, not peripheral. These frameworks pose an important question for AI-powered educational tools: do they support participation and community learning, or do they reinforce an individual, efficiency-driven model aligned with Sfard's acquisition metaphor?

To explore this, I examined three popular free tools—[MagicSchool AI](#), [Diffit](#), and [Eduaide.AI](#). Each highlights a different sociocultural gap and suggests a possible path for redesign.

Tool 1: MagicSchool AI

MagicSchool AI offers over 80 tools for lesson planning, differentiation, assessments, and teacher communication. Although the platform describes itself as a “thriving community of educators,” its design focuses entirely on individual teachers. Each tool, whether for lesson plans, rubrics, or assessments, asks for grade level, topic, and standards, but never for what teachers in the same school are working on together, the shared problems of practice they are exploring, or how professional knowledge might be built collectively. Even its co-planning feature involves teachers creating materials separately rather than observing, analyzing, or reflecting together on student thinking.

This design reflects an acquisition-focused view of teacher learning: knowledge is seen as something created for individuals and measured by hours saved, rather than something cultivated through participation in a community of practice. The lack of mechanisms for shared observation or joint reflection is especially significant for Title I teachers, who already face structural barriers to accessing collaborative PD models like Lesson Study (Lewis & Hurd, 2011).

Redesign Direction: A more sociocultural grounded version of MagicSchool would reimagine its PD module as an asynchronous Lesson Study environment. Instead of producing isolated materials, teachers would form cross-school inquiry groups, explore shared questions about student thinking, utilize AI-generated observation protocols, and participate in structured post-lesson reflection. This redesign directly addresses the platform's current individualistic approach by framing teacher learning as community participation, aligning with Lave and Wenger's idea of legitimate peripheral participation and shifting the tool toward a participation-based view of learning.

Tool 2: Diffit

Diffit is widely used for differentiation and multilingual support. It converts any text into leveled versions, provides translations in 60+ languages, and generates vocabulary and comprehension tasks. Although it claims to support multilingual learners "beyond translation," its core logic treats the home language as a temporary scaffold to be removed once English proficiency improves. The tool simplifies English texts to match the learner's level, implicitly framing the learner as lacking English rather than possessing rich bilingual and community knowledge. Even external descriptions, such as Common Sense Education's (2023) review, position multilingual learners as "struggling readers," reinforcing a deficit orientation.

This design conflicts with sociocultural viewpoints that see learners' linguistic repertoires as valuable assets and starting points for understanding. By emphasizing English as the main source of knowledge, Diffit unintentionally sidelines the community knowledge that Gutiérrez (2008) argues should be the foundation of learning environments.

Redesign Direction: A socioculturally aligned redesign would take a syncretic approach, starting with learners' home-language knowledge and incorporating real community contexts into the materials. Instead of simplifying English texts, the tool would create bilingual, culturally relevant resources where both languages contribute equally to understanding. This change redefines multilingual learners not as lacking but as having valuable linguistic and cultural assets that influence learning.

Tool 3: [Eduaide.AI](#)

Eduaide.AI offers over 100 resource types and bases its outputs on research-backed instructional strategies through an internal Knowledge Graph. However, the “research base” it references is almost entirely cognitive—Bloom's taxonomy, the 5E model, direct instruction, standards alignment—while sociocultural theories, communities of practice, and equity-by-design are missing. Consequently, the tool produces technically aligned materials that overlook who the learners are, what they know from their communities, and how their cultural practices influence learning.

This omission highlights a limited view of learning as an internal, individual process rather than a sociocultural one. Educators using the tool may think they are creating evidence-based instruction, yet the lack of sociocultural frameworks risks making their designs disconnected from learners' lived experiences and community knowledge.

Redesign Direction: A more thorough design would include a necessary sociocultural layer before creating any instructional materials. The tool would prompt educators to identify the learner community, their existing knowledge, and what genuine participation would look like in that context. These inputs would significantly influence the outputs, making sure that the design starts with community knowledge rather than abstract cognitive models. This aligns with Gutiérrez and Jurow's (2016) call for equity-by-design and places learning within real social contexts.

Why I Am Choosing MagicSchool AI for the Final Project

I choose MagicSchool AI because its main limitation is immediately clear: everything is designed for one teacher working alone. Each tool asks for grade level, topic, and standards, but nothing considers what teachers in the same school are trying to understand together or what shared questions they have about students' thinking. Even the co-planning feature has teachers creating materials separately, rather than truly collaborating or analyzing student work side by side. This makes it harder for teachers to develop shared professional knowledge, especially in schools where collaboration time is already limited.

This is also why the redesign feels both meaningful and realistic. A platform that reaches millions of teachers could easily support something like asynchronous Lesson Study, where teachers explore a common practice issue, analyze student thinking, and reflect together—even if they're in different schools or have different schedules. It's one of the few professional development structures that consistently improves instruction, but it's also the hardest to implement in real schools because teachers often lack the time or flexible schedules to meet. A platform like MagicSchool, which already connects with millions of teachers, could make Lesson Study possible in ways schools can't: asynchronously, across schools, and with AI-powered

observation tools. This shift would transform MagicSchool from a simple time-saving tool into a community-driven learning platform.

MagicSchool is also the most practical choice for the final project. It is free, easy to explore, and its limitations are clearly visible in the interface. The redesign is realistic because Lesson Study already has a clear structure that can be translated into features. And the assessment plan is straightforward—I can see how teachers' talk, participation, and attention to student thinking evolve over time. Overall, it provides a concrete, defensible, and equity-focused direction for the final project.

References

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